How are mass extinctions shaping biodiversity?

Confirmation Review

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Part I

Abstract

Mass extinction events are known to generate profound reorganisations of the global biota. From a macroevolutionary perspective, extinction events are destructive for the lineages that die out, but generative for survivors that are offered ecological opportunity in the form of vacant niches. Adaptive radiation theory suggests that these surviving lineages undergo rapid taxonomic and morphological diversification. The diversification of placental mammals after the Cretaceous-Palaeogene (K-Pg) mass extinction has been traditionally viewed as an archetypal adaptive radiation, with the extinction of the non-avian dinosaurs considered the catalyst for the rapid rise in eutherian diversity (placental mammals and extinct relatives). However, the last decade of discoveries in the fossil record and advances in comparative methods have revealed conflicting evidence surrounding the role of the K-Pg extinction in the taxonomic and morphological diversification of eutherians. Recent studies have suggested an ecological radiation may have begun prior to the K-Pg event, while others indicate significant diversification occurring afterward. These discrepancies in findings are likely rooted in contrasting methodologies when investigating morphological diversity (hereafter termed disparity). The central research question of this project is how different methodological choices, such as the type of morphological data, the phylogenetic tree used and various downstream metrics and models applied, influence our understanding of the effects of the K-Pg event on eutherian disparity. To investigate this, I aim to develop a pipeline that iterates through an extensive number of parameter combinations, reflecting the wide-ranging choices available to researchers studying disparity. Once completed, this pipeline will offer a comprehensive reproducible framework for studying the effects of mass extinctions on disparity, which will elucidate how the discrepancies in adaptive signals during the K-Pg event in previous studies may have been a result of contrasting methodological choices. Furthermore, the pipeline can be a useful tool for the wider community to be applied to other clades and extinction events.

Part II

Literature Review

Lit review needs following structure: - What are big evolutionary questions we want to answer? For example, how do animals evolve across a mass extinction? - Inferring evolutionary events with PCM's/morphology -¿ what are the main methods people use. Vast quantities of data have allowed phylogenies to be easily constructed. - Most famously, Raup 1966 coiling (Raup, 1966) - The vast quantities of datasets available. - Phylogenies are growing larger - Disparity -¿ what can it be used to infer - used to infer differences in clade morphospace (are some clades bigger in morphospace than others, more dispersed etc.) - Used to look at patterns through time (DTT)

- However, disparity is not prone to major pitfalls and many of the methodologies need to be made better. - Ancestral state estimation -¿ why we use it? lack of fossils etc. Challenges we face when using PCMs. - describe the methods - continuous -DONE - discrete -DONE - challenges of ace - broad uncertainty, particularly at deeper nodes - using point estimates (cite paper about incorporating uncertainty in macroevo studies) - transition rates in discrete characters and the Darwinian uncertainty paradox - phylogenetic signal - need for fossils (goswami phenomics paragraph, keating, hunt and slater), altho Ane (2008) argues that our confidence interval does not get much lower even with increased sample size. - model misspecification - basically they are shit and often create issues. - they can only recover morphologies that are contained within the observed morphospace - silvestro paper. - perhaps need examples of where ancestral states have proved wrong and how they are vulnerable to model misspecifaition - then say about disparity? are ace estimates carried forward in disparity analyses sensitive? - pre-OASE vs post-OASE.

1 Mass extinctions

1.1 What are they

Extinction is a common process throughout Earth history; 99% of species that have ever existed are now extinct (Overholt, 2003). However, extinction rates are not constant and occur in pulses across background taxonomic turnover (Raup and Sepkoski, 1982). Extinction events that wipe out over 75% of species are termed 'mass extinctions' and have occurred five times over the last 500 million years (Raup and Sepkoski, 1982). Mass extinctions drive a reorganisation of biodiversity as episodes of mass species turnover (Jablonski, 2001). On a macroevolutionary scale, mass extinctions can be considered both destructive and generative; derailing the evolutionary processes of selection and speciation that occur during periods of background extinction (Gould, 2002), yet triggering adaptive radiations in the aftermath as unoccupied niches are filled by surviving clades (Losos, 2010; Toljagić and Butler, 2013). Therefore, mass extinction boundaries often mark the most profound species composition transitions in Earth history.

1.2 Recovery dynamics

The response from the survivors of a mass extinction event may take several forms. Intuitively, the aftermath of a mass extinction may offer surviving clades ecological opportunity, as they are able to fill the vacant niches left by extinct lineages. Classic adaptive radiation theory would suggest that this stimulates simultaneous taxonomic and phenotypic diversification, which are expected to follow a pattern of early high rates that decrease through time as niches become saturated (Schluter, 2000). However, empirical evidence often finds that peak taxonomic diversity and morphological diversity may be reached at different points in a clade's history, suggesting a discordance between these metrics of diversity (Foote, 1993; Hopkins, 2013; Ruta et al., 2013; Wan et al., 2021). This discordance has been observed across mass extinction events, such as in graptaloids across the end-Ordovician mass extinction, (Bapst et al., 2012) or therapsids across the Permian-Triassic mass extinction (Ruta et al., 2013). It has been suggested, therefore, that functional diversity may be density-independent across mass extinction events (Edie et al., 2018). Therefore, understanding the recovery dynamics of clades based solely on taxic diversity metrics may not encapsulate the full complexity of recovery (Puttick et al., 2020).

2 Morphology, Disparity and Extinction

Disparity, the term used to describe morphological variation, has been widely used to investigate patterns of evolution in extant and extinct clades (Wills et al., 1994). Disparity is often used as a way of describing the multidimensional 'trait spaces' that organisms occupy. Guillerme et al. (2020) outlines three ways these traitspaces can be described: size, density and position. Disparity is often used as a proxy for ecological variation, particularly when the morphological traits are inferred to be correlated to ecological function. Mass extinctions pose numerous scenarios of disparity (Figure 1); intuitively, mass extinctions may be expected to result in a decrease in disparity in tandem with selected removal of taxonomic diversity, as well as suppressed morphological evolution during recovery. Alternatively, when extinction is random, disparity may be maintained. In adaptive scenarios, disparity may increase in opportunistic clades that diversify during the period of ecological opportunity following a mass extinction.

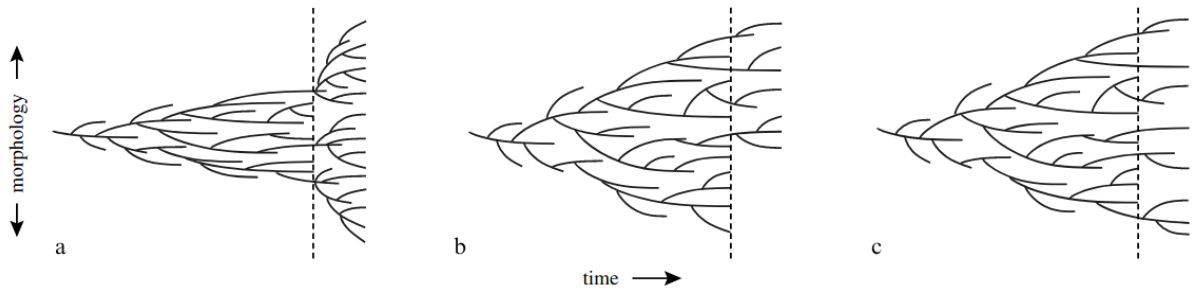


Figure 1: Three scenarios that may occur within a clade across a mass extinction (denoted by dashed line). a) adaptive radiation. b) selective extinction. c) non-selective extinction. Adapted from Grossnickle and Newham (2016).

Idea of extinction selectivity: Devonian events are selective, Permian was non-selective (Korn et al., 2013).

2.1 Extinction selectivity

Mass extinction events can have varying effects on the morphospace of clades. By removing a subset of taxa, the distribution of taxa within the morphospace is transformed. Random extinction will reduce the total number of taxa, but have little effect on variance-based disparity metrics, whereas selective extinction, whereby taxa are more susceptible to extinction due to particular trait combinations, will have a detectable effect on variance-based measures (Foote, 1997; Korn et al., 2013). Korn et al. (2013) proposed several patterns to describe different selection processes: lateral (asymmetric selection that trims one side of the morphospace), marginal (symmetric selection that removes peripheral taxa) and non-selective (random process independent of the variables defining the morphospace). In contemporary extinctions, this idea of 'extinction space' is also valid (Ali et al., 2023; Matthews et al., 2024).

What do we expect: - disparity to increase across mass extinction according to adaptive radiation theory

2.2 Examples of disparity shifting across mass extinction

Perhaps describe the eutherian mammal issue - no real clue on what really happened here? - mixed evidence of disparity changing across mass extinctions - why is that? - Methodological differences could be the answer?

- One example where animals have radiated after a mass extinction (disparity goes up) - One example where disparity decreases (Cole and Hopkins, 2021).

Can link to Cole and Hopkins (2021).

3 Measuring disparity

3.1 Types of trait data

Morphological data used in disparity analyses usually fit into one of two categories: continuous or discrete. Discrete trait data often uses cladistic datasets originally purposed for phylogenetics. Continuous trait data blah blah.

3.2 Pipeline

Show the pipeline fiugure

3.3 Whether or not to include phylogeny

Can link to ancestral state estimation -¿ is ace confounding or not?

4 Ancestral state estimation

4.1 Application of ASE

Ancestral state estimation (ASE) is the broad method of estimating trait value(s) in a set of common ancestors given a set of observed data for operational taxa of a phylogenetic tree (Schluter et al., 1997; Cunningham et al., 1998; Yang, 2006; Revell, 2024). Trait data may be continuous, such as body size, or discrete, such as diet type or presence/absence of a character. The application of ASE is rooted in phylogenetic comparative methods, but also has applications in linguistics. As put by Swofford and Maddison (1992), 'character state reconstructions can provide a powerful mechanism for studying many facets of the evolutionary process. However, the zeal with which these techniques are sometimes advocated belies the complexity of the problem'.

Intuitively, ASE is appealing for those studying deep-time patterns of disparity. The fragmented nature of the fossil record means that means much of the phylogeny is left unsampled when including only observed taxa. Incorporating a phylogeny into disparity analysis will unavoidably introduce unsampled 'ghost lineages' (Brusatte et al., 2011). A ghost lineage is the hypothetical ancestor that is not observed in the fossil record, but can be inferred to have existed due to the presence of other fossils/extant taxa. This can be best explained as the branch on a phylogeny between two observed taxonomic units (see below)

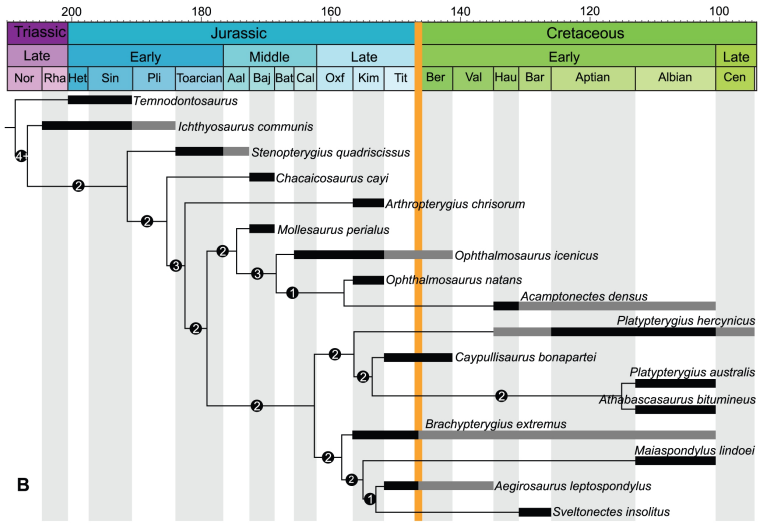


Figure 2: Phylogeny of ichthyosaurs showing observed stratigraphic range for each species (thick lines) and inferred ghost lineages (thin lines) (Fischer et al., 2012).

ASE has been employed as a way of inferring the traits of these hypothetical species based on the traits of their descendents. This fills in the gaps in the fossil record and alleviates the bias associated with uneven sampling across the phylogenetic tree (Friedman, 2009; Brusatte et al., 2011). When using ASE, the estimated traits for nodes of a phylogeny are used as a way of sampling these ghost lineages. As detailed below, modern ASE methods make use of branch lengths to describe the temporal range over which morphological evolution has taken place. Thus, longer ghost lineages can increase uncertainty in ASE estimates, which highlights the need for fossil sampling. The importance of accounting for ghost lineages has been stressed, since it can impact our understand of the rate of morphological evolution. Ignoring ghost lineages may find abrupt shifts in disparity that may be misleading of a more gradual accumulation that has been missed due to incomplete fossil sampling. Therefore, using ASE tends to smooth the disparity curves through time. Look at Halliday. guillerme etc for how to incorporate these aced estimated into dtt - different binning methods, time slicing etc.

An additional requirement of ASE is the model specified. This model specification varies between discrete and continuous characters.

(Friedman, 2010).

The fragmented fossil record can make sample size at deeper points in the phylogeny low, which may necessitate ASE.

4.1.1 Discrete characters

Discrete characters are typically analysed using parsimony or likelihood approaches, although likelihood methods are generally preferred. Likelihood approaches, such as Bayesian inference or Maximum likelihood, optimize solutions based on the tree topology, branch lengths and the model of character evolution. Discrete

characters are modelled using a continuous time Markov chain process, known as the Mk model (where k = number of possible states) (Pagel, 1994; Yang, 2006). This process is specified by a $k \times k$ matrix, $\mathbf{Q}$, which gives the instantaneous transition rates (q) between states for each character.

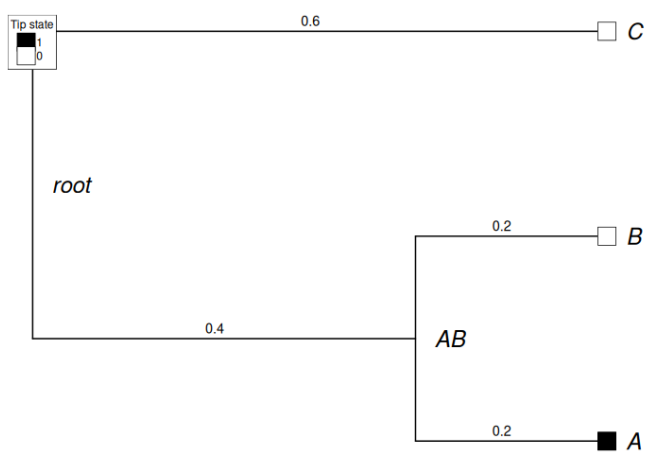
$$\mathbf{Q} = \begin{bmatrix} -q_{0,1} & q_{0,1} \\ q_{1,0} & -q_{1,0} \end{bmatrix} = \begin{bmatrix} -0.5 & 0.5 \\ 0.5 & -0.5 \end{bmatrix}$$

There are a number of Mk transition rate models that can be fit to data. The 'equal-rates model' (ER) is the simplest and estimates just one parameter, since it assumes all transitions occur with equal probability, equivalent to the Jukes-Cantor model of molecular evolution. The assumptions of the ER model can be eased with the 'symmetric' model (SYM): rate of change between any two character states is equal in the forwards and backwards direction, but may differ between different pairs of states. The 'all-rates-different' (ARD) model allows for a different transition rate in each direction, includes a total of $k \times (k-1)$ parameters to estimate.

This likelihood approach makes use of branch lengths, which is why it has become the preferred method of ASE compared to previous methods, such as parsimony. The probability of each transition at any given interval of time (t) can be calculated by

$$\mathbf{P} = \exp(\mathbf{Q}t)$$

So if we want to calculate probability of each transition at node AB (ancestor of A and B) $t = 0.4$.



$$\mathbf{P}_{t=0.4} = \exp(\mathbf{Q} \times 0.4) = \begin{bmatrix} 0.835 & 0.164 \\ 0.164 & 0.835 \end{bmatrix}$$

One further probability to consider is the 'root prior', which is essentially whether the root node of the

tree was in state 0 or 1. This is typically set with a flat root prior $\pi_0 = \pi_1 = 0.5$.

The most popular method of ASE is *marginal* estimation, which involves proceeding from node to node on the phylogeny, and, at each node, computing the probability of observing the tip data of our tree conditioned on fixing the node we've visited to each one of the set of distinct values of our trait. If we were to compute the probability that the root was in state 0,

For example, if we were to calculate the probability of the root node was in state 0:

$$\begin{aligned} P(\text{root} = 0) = \pi_0 \times P(0|0, t_{AB}) \times P(1|0, t_A) \times P(0|0, t_B) \times P(0|0, t_C) \\ + \pi_0 \times P(1|0, t_{AB}) \times P(1|1, t_A) \times P(0|1, t_B) \times P(0|0, t_C) \end{aligned}$$

This is repeated for $P(\text{root} = 1)$ and then the likelihoods are rescaled by their sum so they add up to 1.0. In marginal estimation, this is calculated node by node. Since this can become extremely complicated as the phylogeny increases in size, in practice functions use algorithms such as Felsenstein's pruning algorithm (Felsenstein, 1981).

4.1.2 Continuous characters

Continuous trait evolution is typically modelled under a Brownian motion (BM) model of evolution (Felsenstein, 1985; O'Meara et al., 2006). Brownian motion models trait evolution as a directionless, stochastic process under which continuous characters undergo changes along the branches of a phylogeny. These evolutionary changes come from a Gaussian distribution with a mean of 0 and variance proportional to the product of the rate of evolution and time elapsed ($\sigma^2 t$). Therefore, the expected amount of trait divergence between any two taxa is proportional to the amount of shared evolutionary history according to the phylogeny.

Under BM, the vector ($\mathbf{x}$) of observed tip trait values follows a multivariate normal (MVN) distribution and has an expected value that is equal to the root state x_0.

$$\boldsymbol{x} \sim \text{MVN}(x_0,\, \sigma^2 \boldsymbol{C})$$

$\boldsymbol{C}$ is an variance-covariance matrix of N dimensions where the diagonal elements $\boldsymbol{C}_{ii}$ accelerated the tree height from root to tip i and $\boldsymbol{C}_{ij}$ is the height above the root of the most recent common ancestor of i and j. When estimating ancestral states under BM, the trait values of the internal nodes are also distributed within this MVN distribution. Therefore, the $\boldsymbol{C}$ matrix can be expanded by one column and row for each node (except the root). During Maximum Likelihood, the x_0 and σ^2 parameters are estimated as the values that maximize the likelihood of observing the tip vector. Internal node values can be estimated using the conditional expectations of this distribution, i.e. the trait value at an internal node is calculated as a weighted average of the observed tip values, determined by the amount of shared evolutionary history (given by $\boldsymbol{C}$) and the node's position in the tree.

4.2 Challenges of ASE

'Character state reconstructions can provide a powerful mechanism for studying many facets of the evolutionary process. However, the zeal with which these techniques are sometimes advocated belies the complexity of the problem' (Swofford and Maddison, 1992).

4.2.1 Uncertainty in estimates (and how to deal with it)

A critical issue within ancestral state estimation is that estimates tend to have broad uncertainty (Schluter et al., 1997). This uncertainty is not uniform across the tree: it typically increases towards root of the tree, where direct information from observed tip data becomes increasingly diffuse (Gascuel and Steel, 2014). For discrete characters, this uncertainty manifests as inconclusive scaled likelihoods, i.e., state 0 = 0.51, state 1 = 0.49. For continuous characters, uncertainty is wide confidence intervals. Oftentimes, these confidence intervals are so wide that it may be counter-intuitive to even bother with ASE (Martins, 2000).

How this uncertainty in estimations should be dealt with is equivocal. Despite the encouragement for uncertainty to be carried forward in comparative analyses (Silvestro et al., 2015), most computational implementations of ASE return point estimates. For continuous characters, this will be the median value of the confidence interval and for discrete characters, it will be the most probable state according to a threshold (see later). By ignoring uncertainty, point estimates can lead to overconfidence in results and erroneous inferences in downstream analyses (such as down the disparity pipeline), all while obscuring potentially 'accurate' uncertainty.

For discrete characters, there is some level of subjectivity in how to collapse the states into the most probable based on their scaled likelihood. It is common to have a threshold implementation, whereby the user can set at what threshold of likelihood states are collapsed to a point estimate, or left as uncertain, which is usually denoted as '0/1' for example. However, these thresholds are often arbitarily set with no guidance of what is the 'correct' value (Smith and Donoghue, 2022; Casali et al., 2023). Much of the subjectivity relates to the trade-off between permissiveness and conservativeness; higher thresholds are likely too conservative and so too many states are left uncertain, perhaps missing true signals. Whereas lower thresholds may be too permissive and can lead to erroneous inferences.

4.2.2 Sensitivity to fossil sampling

Given that uncertainty is often high in ASE, how then can estimates be improved? Ané (2008) demonstrates that ASE is *statistically inconsistent*, meaning that even if the sample size of extant tips is infinite, estimates are not expected to converge on the true value. Instead, the inclusion of fossils can improve estimates, due to the shorter evolutionary path to the root of the tree (Finarelli and Flynn, 2006; Finarelli and Goswami, 2013). Fossils can be used as priors on nodes, or simply added to the tree as tips. Fossils become more important when model assumptions are violated (Puttick, 2016).

The necessity of fossil inclusion extends to disparity analyses too. Mitchell (2015) found that analyses of bird disparity were underestimated when using extant-only datasets.

Adding fossils will be able to level this mean out. A similar situation could occur with a mass extinction event. Increasing the number of extant taxa does not improve ase estimates (Ané, 2008).

4.2.3 Model dependency

Another key challenge with ASE is its sensitivity to model misspecification. For continuous characters, this is particularly pertinent when traits have evolved under non-BM (Revell, 2024). For example,

These model violations can occur in several ways. For example, if a clade evolved under a BM trend model (a moving average), current implementations of ASE assume a fixed mean, which, if given only extant taxa, will find a mean skewed towards the tips rather than root. Both Webster and Purvis (2002) and Finarelli and Flynn (2006) were unable to recover accurate ancestral estimates of body size in planktonic foraminifera and mammals, respectively, due to a trend of increase through time. Since ancestral states are limited to what is observed, and we know that trait distributions can change dramatically through time, fossils provide the most reliable evidence of the trait distribution at that point.

5 Detecting shifts in disparity

6 Aims and objectives

Knowledge gap

Part III

Thesis plan

7 Chapter One: What is the effect of ancestral state estimations on disparity?

7.1 Introduction

- Morphology can tell us interesting patterns about evolutionary processes.
- The combination of morphological datasets (using fossils) and phylogenies means we can infer these processes through deep time

- This is particularly useful when studying major evolutionary events (such as a mass extinction)
- However, the fossil record is incomplete and this restricts our ability to look at deep time patterns of morphology.
- Many studies have made use of ancestral state estimation (ASE) to fill in gaps left by 'ghost lineages'. We know some species must have existed between root and tips but we don't know what they looked like.
- Particularly has been implemented in studies looking at disparity, which describes the multidimensional 'morphospace' in which taxa occupy.
- ASE can be used to estimate the character state values (both continuous and discrete characters) at nodes on a phylogeny, using observed trait data (from living and fossil specimens).

7.1.1 Why does this matter?

- ASE estimates are often shit and a 'glorified average'. But, what can we do with them?
- Erroneous estimates can mislead values of disparity and therefore macroevolutionary inferences.

7.2 Methods

7.2.1 Simulations

- I simulated a set of 100 trees using `treats` (Guillerme, 2024). These trees were simulated with birth-death parameters as $\lambda = 1.0$, and $\mu = 0.7$. This leaves trees with a range of 50-200 fossil tips and 50 extant tips.
- I simulated discrete and continuous trait data using `map.traits`. This effectively maps the 'true' evolutionary history across the phylogeny (tips and nodes have states mapped).
- For discrete traits, I simulated trait data across each tree (100 characters, 85% binary, 15% 3-state) using Mk model with 3 different transition rates: fast = 0.9, medium = 0.1, slow = 0.01.
- For continuous traits, I simulated BM and OU model trait data (50 characters).

7.2.2 Fossil sampling

- I then removed all node states from the matrices, leaving just tip states.
- In order to understand how ASE accuracy is affected by fossil sampling level, I pruned the fossil tips at 5 levels, preserving the root to maintain tree height across each level:
- **Living**: only extant tips are retained

- **Low**: 5% fossils retained
- **Medium**: 15% fossils retained
- **High**: 50% fossils retained
- **All**: All fossils retained

7.2.3 ASE

- I then ran ASE on these matrices using `multi.ace`, with three different parameters (for discrete states).
- **Relative majority rule**: the ancestral state estimated are all the ones that are have a scaled likelihood greater than the maximum observed scaled likelihood minus the inverse number of possible states:

$$\text{estimated state} \geq \left(\max(\text{likelihood}) - \frac{1}{n_{\text{states}}}\right)$$

  If more than one state is over this threshold, it is recovered as uncertain e.g., (0/1).
- **Strict majority rule**: Setting threshold parameter to a set value will select the ancestral states estimate with a scaled likelihood equal or greater than this designated value (in this case I did 0.5).
- **No majority rule (sampling)**: `multi.ace` has a `sample = x` parameter, whereby you can return x matrices, in which states returned are by the scaled likelihood of each state appearing at each node. In this case I used sample = 10 for fast computation (in future I will do 100/1000).
- Ran `multi.ace` on continuous characters using BM as model.
- I will also do post-ordination ace to see how that compares to pre-ordination ace.

7.2.4 Raw distance comparison between true vs ace

- I then used a Hamming distance metric to compare corresponding nodes between the 'true' vs 'aced' matrices.
- Distances were calculated only for nodes that were present in all fossil sampling levels to allow for comparison.
- The distance assigned a penalty of 1 for any wrong estimated state and also for uncertainty (i.e 0/1).
- For continuous traits, I firstly scaled each matrix and then measured the euclidean distance between true vs ace for each corresponding node.
- I also looked at the distance at just the root, to see how well ace recovers true value at deepest node.

7.2.5 Distance comparison between ordinated true vs ace

- I then ordinated the true matrix (distance matrix then PCO for discrete, PCA for continuous), then projected the aced matrices onto this true matrix.
- I measured the Euclidean distance between corresponding nodes again.

7.2.6 Difference in disparity between true vs ace

- from the ordinated matrices, I calculated (will choose which metrics, at the moment sum variance and ranges and average nearest neighbour distance) disparity and calculated the difference between true vs aced.

7.3 Results (or what I expect to find)

7.3.1 Raw distances

- Slower transition rates = ASE accuracy improves
- Increasing fossil sampling improves ASE accuracy (for discrete traits), particularly at deeper nodes.
- Using sample method of ASE improved estimate accuracy
- Using sample method is relatively insensitive to low fossil sampling and finds high accuracy at all fossil sampling levels
- At the root node, the impact of fossil sampling becomes more pertinent i.e., higher fossil sampling improves estimates
- For continuous traits, ASE accuracy was much better for BM traits than OU
- Estimates of continuous traits stay relatively constant across fossil preservation levels.
- For continuous traits, using point estimates was more accurate than sample method.

7.3.2 Ordination

- I expect that patterns of distance to true value will correspond to the raw distance pattern (more fossil sampling = higher ASE accuracy, sample method = higher ASE accuracy)
- I expect that fast transition rates will exhibit clustering at the centre of the morphospace, particularly with the default multi.ace parameter, due to the high amount of uncertainty recovered at deeper nodes. I think that the sample method of ASE will alleviate this artefact by removing this uncertainty.

7.3.3 Disparity

- Preliminary results have found that using sample ASE method recovers high accuracy for all transition rates.
- Using the default ASE method recovers much lower accuracy, particularly with low fossil sampling.
- I expect to see different results for different disparity metrics. For example, metrics that focus on density will be able to identify these artifacts of clustering due to uncertainty in node estimates, which variance based metrics will miss.
- This will highlight the need for multiple different disparity metrics to be used in disparity analyses

8 Chapter Two: Measuring changes in trends in disparity through time

9 Chapter Three: Eutherian disparity over K-Pg mass extinction

Part IV

Summary of research achievements

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